

International perspective: market design, adequacy mechanisms, and the role of storage and demand response

Panel: Firm capacity markets in systems with high penetration of variable energy sources and storage systems — Latin America and beyond

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Resource adequacy is being redefined

Deep decarbonization changes what “firm” means

- ▶ **From peak MW to every hour.** Risk migrates from the demand peak to net-load peaks, steep ramps and multi-day low wind / solar / hydro events — MISO now sees grid emergencies in all four seasons
- ▶ **Adequacy is now multi-dimensional.** Firm capacity (MW), firm energy (MWh across weeks) and flexibility become distinct products that markets must value
- ▶ **Storage and DR change the question.** They are adequacy providers — but duration-limited and state-of-charge constrained, so accreditation becomes the central design problem
- ▶ **The “missing money” problem gets worse.** High VRE shares depress energy prices exactly when investment in firmness is most needed — hence the global (re)turn to adequacy mechanisms

Market design question of the decade: how to buy firmness in a system where few resources are firm by themselves

MW — firm capacity

Cover the net-load peak with resources that show up when it matters.

MWh — firm energy

Ride through droughts, fuel constraints and multi-day VRE lulls.

MW/min — flexibility

Follow steep ramps and volatility as VRE and large loads grow.

One goal, many instruments

A taxonomy of how systems buy firmness

Design family	Where	Key features
Long-term contract auctions + capacity reserve	Brazil (LEN / LRCAP)	Contracts as the financing backbone; dedicated reserve-capacity (“lastro”) products since 2021
Firm energy obligations + reliability options	Colombia	Firm-energy product (OEF) with a strike price that hedges consumers during scarcity
Capacity payment based on verified generation in critical hours	Mexico	Reliability contribution measured by generation verified during a fixed number of critical hours per year, identified by the system's capacity margin; more predictable than spot-price-based periods
Centralized capacity markets	PJM, MISO, ISO-NE, GB, Italy, Belgium, Poland	Capacity obligations auctioned years ahead; accreditation via ELCC / de-ratings
Energy-only + scarcity pricing	ERCOT	ORDC adders and high offer caps; investment rides on volatile scarcity margins

The design choices that actually matter

Five questions that separate adequacy mechanisms

1 • What is the product?

Peak MW, firm energy or flexibility? Single or multi-product? The product must match the system's dominant risk — capacity, energy or ramping.

2 • Who counts as firm?

Accreditation: average vs marginal ELCC, de-rating by duration, storage and hybrid rules. Small methodological choices move billions.

3 • Who buys, how far ahead?

Central auctions vs decentralized obligations; lead times (1–7 yr) and contract lengths (1–15+ yr) determine financeability of new entry.

4 • What if it under-delivers?

Ex-post settlement: performance penalties, reliability-option paybacks, firm-energy verification — incentives must bind exactly at the critical hours.

5 • Who pays — and is it politically durable?

Cost allocation to load and tolerance for price spikes decide whether a design survives its first crisis. Recent history: PJM caps negotiated after a 10× price jump; Texas shelved its PCM; Colombia is debating the charge itself.

Brazil: auctions, now for batteries

Capacity reserve auctions (LRCAP) now reach storage

- ▶ **Auction-based expansion.** ~100 GW contracted since 2005 at ~US\$50/MWh on average; since 2021, capacity reserve auctions (LRCAP) procure firmness (“lastro”) paid by all consumers — a de facto separation of reliability and energy products
- ▶ **First battery-only LRCAP — December 2026.** 15-year contracts with delivery from August 2028; two products (national-content and open); ~2-3 GW of estimated demand and ~R\$10 bn in first-round investment
- ▶ **Design choices under debate.** Batteries separated from thermal capacity products while storage regulation matures; ANEEL rules, grid-services value stacking and delay risk remain open
- ▶ **Broader reform context.** Market opening (retail liberalization approved for 2028) proceeds in parallel — capacity products must coexist with decentralized contracting

Lesson: auctions create a bankable route to market for storage even before full market rules exist — LATAM’s classic recipe, applied to a new technology. On the other hand, because Brazil has always been an auction-driven country, this tends to keep “delaying” the shift toward — and the development of — a genuine short-term market

Dec 2026

first battery-only capacity reserve auction

~2-3 GW

estimated demand for the first battery LRCAP

R\$ 10 bn

potential investment unlocked in the first round

Colombia: the firm-energy product

The reliability charge (Cargo por Confiabilidad) under stress

- ▶ **A LATAM innovation (2006).** Firm Energy Obligations (OEF) with reliability options: generators sell verified firm energy and pay back above the scarcity price — a built-in consumer hedge
- ▶ **How it is paid.** A plant that clears the firm-energy auction earns the reliability charge (Cargo por Confiabilidad) — a fixed payment for its verified firm capacity. The last auction cleared at 22 USD/MWh
- ▶ **A cap for consumers.** Demand always pays this charge; in return, whenever the spot price exceeds the strike price, generators pay back the excess — so consumers never pay more than the strike price and are shielded from price spikes
- ▶ **Designed for hydro risk.** The product targets El Niño droughts: multi-week energy firmness rather than peak MW — exactly the multi-day risk VRE-heavy systems elsewhere are now discovering
- ▶ **The gap.** Batteries and DR still lack a clear route to sell firm energy — accreditation of duration-limited resources is the next frontier

Lesson: Colombia's firm-energy product anticipated by 15+ years what others now call multi-day adequacy. Firm energy is the right product where the binding risk is energy, not capacity

OEF

firm-energy obligations with reliability options since 2006

2029-30

expansion auction under way — storage incentivized

El Niño

the design risk: droughts, not peak demand

Mexico: the criticality trigger

Firmness measured over the 100 tightest hours of the year

- ▶ A **“criticality trigger.”** Instead of waiting for realized price spikes, critical hours are approximated by the system’s capacity margin — the gap between available supply and demand
- ▶ **The 100 tightest hours.** A fixed number of hours each year (the ~100 tightest) is designated as the critical period; a generator’s firm capacity equals its availability / actual generation during those hours
- ▶ **Predictable by design (?).** Because periods are set by capacity margin — not realized spot prices — they are more predictable and give more frequent signals for agents to adjust strategy
- ▶ **The trade-off.** Standardized critical hours are only correlated with true scarcity — less directly tied to the spot prices

Lesson: standardizing the critical hours trades pinpoint accuracy for predictability (standardized periods cap generator exposure)

~100 hrs

the “tightest” hours each year that define the critical period

Margin

criticality is triggered by the system capacity margin, not spot price

PJM: the capacity crunch

Two consecutive auctions at the price cap — and still short

- ▶ **Prices at the administrative ceiling.** 2025/26 cleared at \$269.92/MW-day (~10× the prior auction, a \$14.7 bn bill); 2026/27 and 2027/28 cleared at the FERC-approved cap of \$333/MW-day
- ▶ **Demand is the story.** PJM forecasts 32 GW of peak-load growth through 2030 — 94% from data centers. The reliability requirement rose 5+ GW in one year; the target reserve margin was lifted to 20% on winter risk
- ▶ **Supply is not keeping up.** Only ~0.8 GW of new capacity cleared between the last two auctions; 2027/28 closed 6.6 GW short of the requirement — the first RTO-wide shortfall
- ▶ **Accreditation reform repriced firmness.** Marginal ELCC moved billions; DR was re-rated 69% → 92% with 24-hour availability; batteries' capacity returns rose ~17.5% auction-on-auction
- ▶ **Emergency backstop auctions.** A Jan/2026 federal–state “Statement of Principles” proposes a one-time auction with 15-year capacity contracts to underwrite ~\$15 bn of new dispatchable generation — costs borne 100% by data centers via take-or-pay. PJM has proposed its own pre-emptive backstop
- ▶ **Sound familiar?** Auctioning long-term contracts to backstop adequacy is the LATAM playbook (Brazil's LRCAP, Chile's supply auctions) — now proposed in the world's flagship capacity market

A capacity market can reveal scarcity — it cannot, by itself, build through an interconnection queue

\$530/MW-day

where 2027/28 would have cleared without the cap

6.6 GW

shortfall vs the 152 GW reliability requirement

94%

of 32 GW load growth to 2030 comes from data centers

ERCOT: no capacity market needed?

Scarcity prices and new reliability services, no obligations

- **Energy-only by choice.** ORDC scarcity adders and high offer caps remunerate firmness; the \$1 bn Performance Credit Mechanism was legislated in 2023 — and shelved in December 2024.
- **A storage boom on market signals alone.** More than 16 GW of batteries operate on energy and ancillary-service arbitrage — the largest US fleet, built without any capacity payment.
- **The toolkit keeps evolving.** Real-time co-optimization of energy and reserves, a new Dispatchable Reliability Reserve Service, and ADER pilots bringing aggregated DER / DR to market.
- **The trade-off.** Price spikes are the investment signal — so reliability “insurance” stays implicit and depends on political tolerance for volatility.

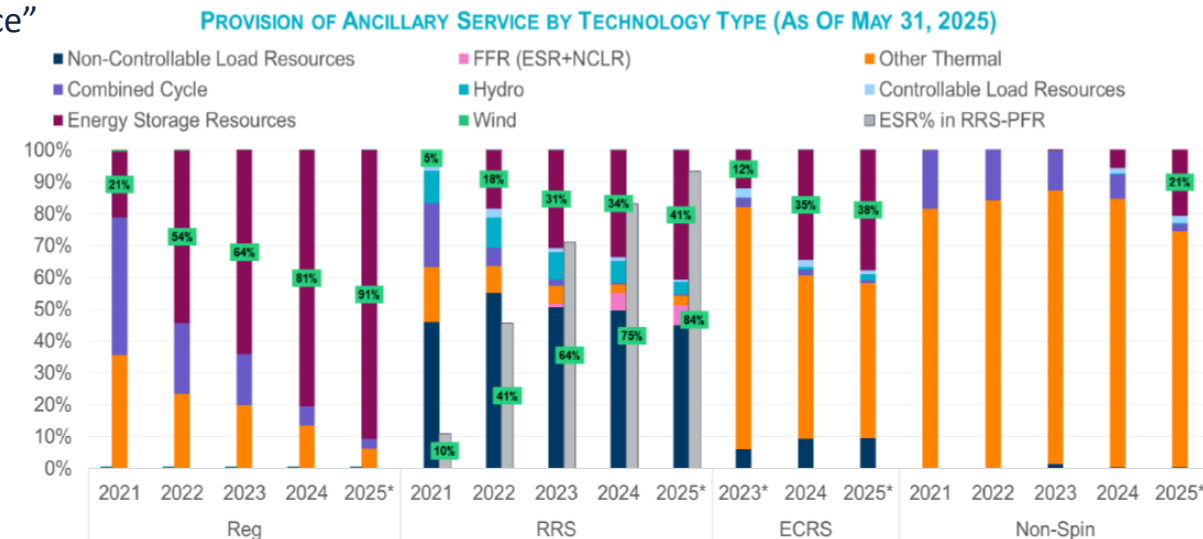
16+ GW

batteries online in ERCOT — vs < 0.5 GW in PJM

\$0

paid through a capacity market — there is none

*Lesson: sharp price signals can mobilize storage and DR at scale
— the question is whether society accepts the volatility that comes with them.*



Key Takeaways



There is no single right mechanism — but there are wrong products: designs that pay pure MWh without pricing duration, energy firmness and flexibility will misfire in VRE-heavy systems



Adequacy mechanisms are here to stay: even “energy-only” systems keep adding reliability services



Storage and DR gain participation rights in every framework



Accreditation is key: how firm capacity is credited decides which resources get paid and whether the mechanism actually delivers reliability

THANK YOU!

Any questions?

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